USAT Project Podium

Metabolic Testing and Data Analysis Update

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Project Podium Overview

Strength	Description
Daily Training Environment	• Immersive, year-round environment integrating swim, bike, and run sessions with strength, recovery, and tactical skills work. • Small, focused athlete group allows for individualized programming and daily coach–athlete interaction. • Emphasis on fostering a competitive yet collaborative training culture that prepares athletes for the demands of the World Triathlon circuit.
Supreme Supporting Partners	• Backed by world-class equipment, apparel, and nutrition sponsors, ensuring athletes have the same tools as Olympic medal contenders. • Access to cutting-edge sport science through USOPC, university partners, and technology providers for performance testing, biomechanics, and data analysis. • Integrated support team including physiotherapists, nutritionists, sport psychologists, and bike mechanics. • Partnerships that offer mentorship, networking, gear, and professional development opportunities.
Scholarships and Support	• Partnership with Arizona State University provides tuition assistance, enabling athletes to pursue degrees while training and competing. • Academic advising and flexible class scheduling tailored to elite race calendars. • Life skills and career development workshops to prepare athletes for post-sport careers. • Proven track record of producing graduates who excel both in sport and in professional pathways.
Facilities and Resources	• Access to world-class training venues in Tempe, AZ, and Park City, UT: Elite Pools, private open water swimming areas, indoor/outdoor tracks, dedicated cycling routes, and altitude training opportunities. • On-site recovery and rehabilitation facilities, including ice baths, compression therapy, and sports massage. • High-tech performance lab for physiological testing, lactate analysis, and power/HR monitoring. • Year-round training in an optimal climate that supports consistent outdoor swimming, cycling, and running.

Competition Overview

➢ Project Podium athletes benefit from unique competitive opportunities spanning domestic, regional, and international levels, enabling them to gain experience, earn ranking points, and develop toward the world stage.

Domestic and Regional Competition

USAT National Championships – Project Podium athletes have excelled at the junior national level, winning the last 5 USA National Championships

Development Pipeline – Provides race opportunities for younger athletes to build skills, confidence, and readiness for international racing

World Triathlon Hosted Events

Continental Cups – Foundation of senior elite international racing, giving athletes the chance to earn valuable World Triathlon ranking points and gain high-level race experience

World Cups – Global-level elite races attracting strong international fields, offering higher points and exposure

World Triathlon Championship Series (WTCS) – Premier annual series featuring the world's top athletes, awarding maximum points outside the Olympics

Other Events

Single Sport Racing (Track Meets, Cycling TT/RoadRacing, etc.) – Project Podium gives athletes the opportunity to compete amongst other collegiate and professional athletes in single sport disciplines. In the case of the track, this can help sharpen the athletes run speed and provide a different stimulus and environment leading into the season.

XTERRA – Project Podium boasts some of the best short course athletes in the off-course triathlon discipline.

Data Analysis Overview

- Parker Spencer has led Project Podium to the cutting edge of data science techniques within Triathlon. The combination of metabolic testing, race simulation, and modelling help provide athletes with specialized training programs and insights over the competition

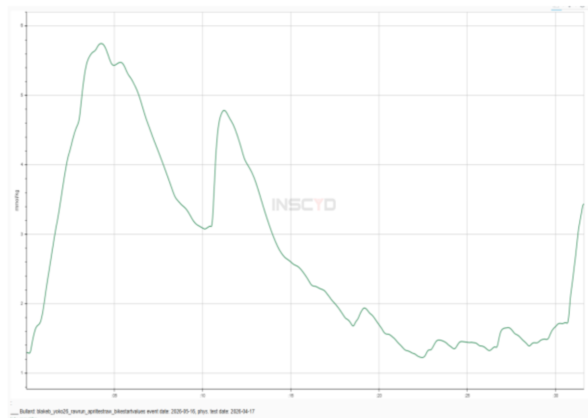
Metabolic Testing

INSCYD

INSCYD Partnership: Partnered with premier metabolic software provider INSCYD. Provides specialized reports for the athlete based on V02 max, VlaMax, and other key endurance markers.

Testing Procedure: Regular testing throughout the year ensures accurate training zones such that workouts are always tailored to the individual athlete's needs rather than a broad group plan.

Race Analysis & Simulation



Historical Race Analysis: Data analysis on historical athlete performance to quantify improvement and compare current development athletes to Olympic Medalists past trajectory.

Individual Race Lactate Simulation: Using the metabolic profile of an athlete in combination with the athlete's race file allows us to simulate lactate levels during the race to identify tactical errors and training needs.

Future Race Modelling

Elite Men — 2026 World Triathlon Championship Series Alghero
2026-05-30 | Alghero, Italy | 55 athletes | 10000 simulations

Breakaway Bias

Det. Rank	Athlete	Country	Win %	Podium %	Top5 %	Top10 %	E[Rank]
4	Vasco Vilaca	Portugal	**23.1**	**49.5**	66.4	87.7	5.0633
1	Miguel Hidalgo	Brazil	7.0	**27.0**	47.4	80.3	6.8461
7	Hayden Wilde	New Zealand	**11.6**	**31.6**	47.4	74.5	7.5613
3	Matthew Hauser	Australia	**11.0**	**30.6**	46.7	74.4	7.5316
12	Alessio Crociani	Italy	10.0	**27.3**	42.4	68.8	8.4847
17	Vetle Bergsvik Thorn	Norway	4.6	**17.2**	31.0	61.0	9.7999
8	Tyler Mislawchuk	Canada	5.3	**16.3**	27.2	50.9	11.8262
13	Oliver Conway	Great Britain	5.0	**15.2**	25.5	49.1	12.1583
11	Luke Willian	Australia	3.7	**13.0**	22.3	46.1	12.7348
8	Csongor Lehmann	Hungary	2.8	9.6	17.6	39.5	14.1785
15	Charles Paquet	Canada	3.7	**11.8**	19.8	39.6	14.471

Race Briefings: Perform analysis on past performances for a given course to prepare athletes for the likely demands of competition and identify competitors well suited for the predicted race conditions

Race Simulation: Developing Monte Carlo Simulation based on decades of triathlon race history to help predict race results. Will help identify when an athlete is ready for the next level of competition.

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Race Overview – WTCS Yokohama

WHO

- Three Project Podium Athletes competed at the WTCS Yokohama – Braxton Legg, Blake Bullard, Reese Vannerson
- These athletes have similar metabolic profiles; however historically Reese Vannerson has been the top runner

WHAT

- All three athletes came off the bike together in the main chase group. In this position, Reese was most likely to have the best performance
- Yet, he struggled on the run and finished 3rd among the group

HOW

- When analyzing the athletes bike data, Reese's file in terms of average power looks very similar to Blake's and Braxton's.
- However, using advanced simulation techniques we model out Reese's Blood Lactate levels and see that he finished the bike far above his threshold. This led to poor run performance

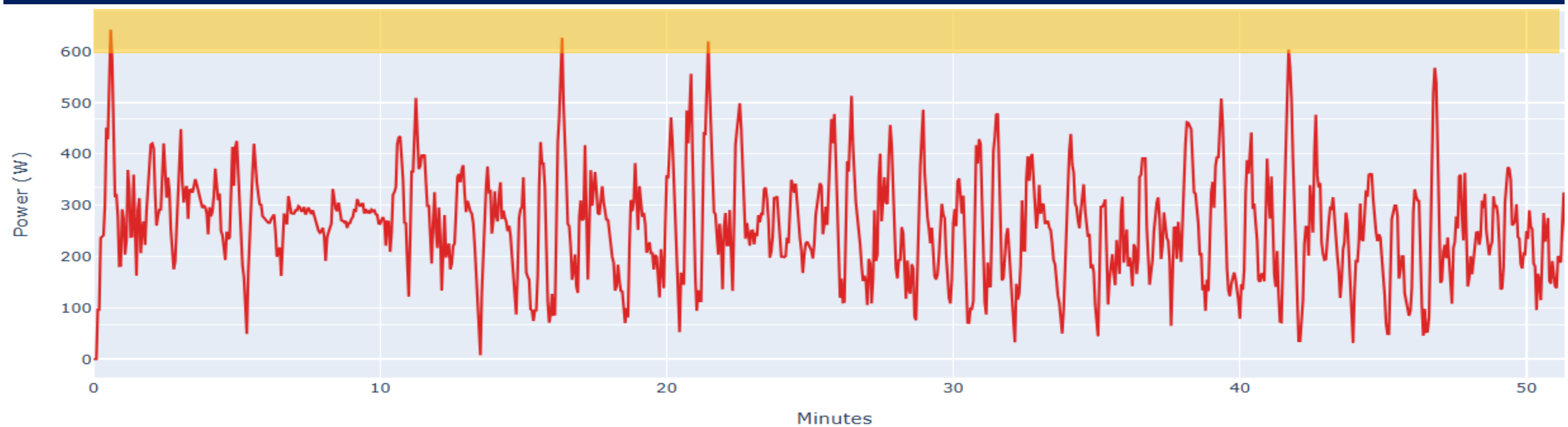
WHY

- This is important because without this data we might have believed Reese to have lost significant run fitness.
- Instead, it's clear that Reese's run was more so affected by the surges on the bike that spiked his lactate. With more bike handling preparation and awareness, we can help the athlete set himself up for their maximum potential on race day.

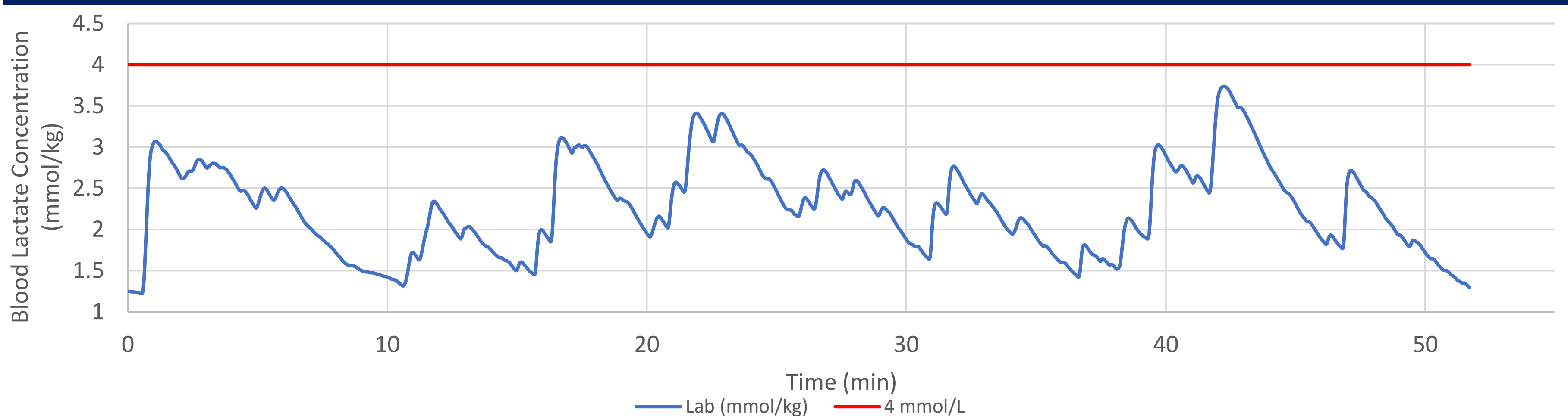
Blake Bullard Yokohama Bike

- Despite being in the breakaway, Blake minimized his surges over 600W once. Then when falling back into the main chase, Blake rode mostly around a medio effort keeping lactate below 4 mmol/kg

Raw Power File



Simulated Blood Lactate Concentration

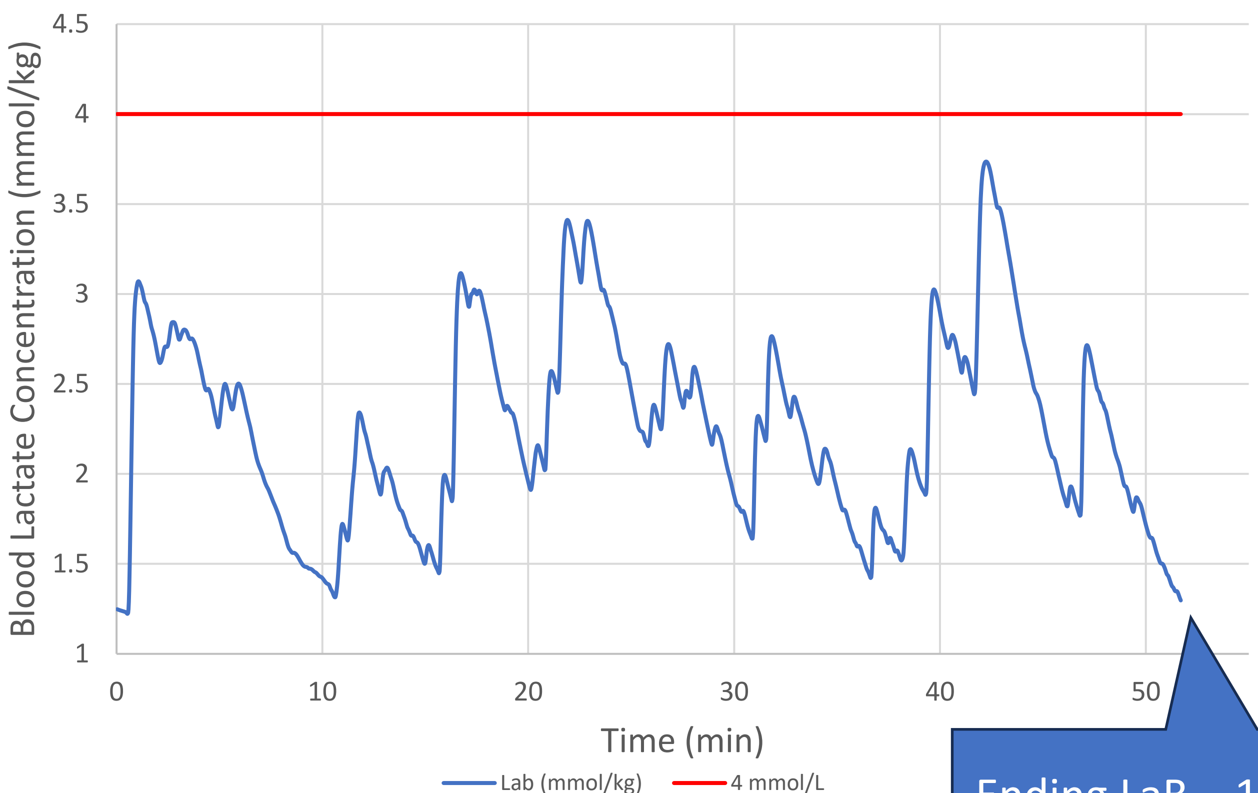


Note: Power file shown with 2s smoothing for better visibility

Blake Bullard Yokohama Bike

- Blake began the bike in a small 7-man breakaway coming out of the water yet had his highest spikes in lactate once he had been dropped and settled into the main chase group. This suggests that Blake needs to continue working on bike skills rather than pure power numbers
- Blake relatively even pacing resulted in a VI of 1.08, always keeping lactate below 4 mmol/kg and even dropped it all the way below 2 mmol/kg before starting the run

Simulated Blood Lactate Concentration



Ending LaB – 1.3

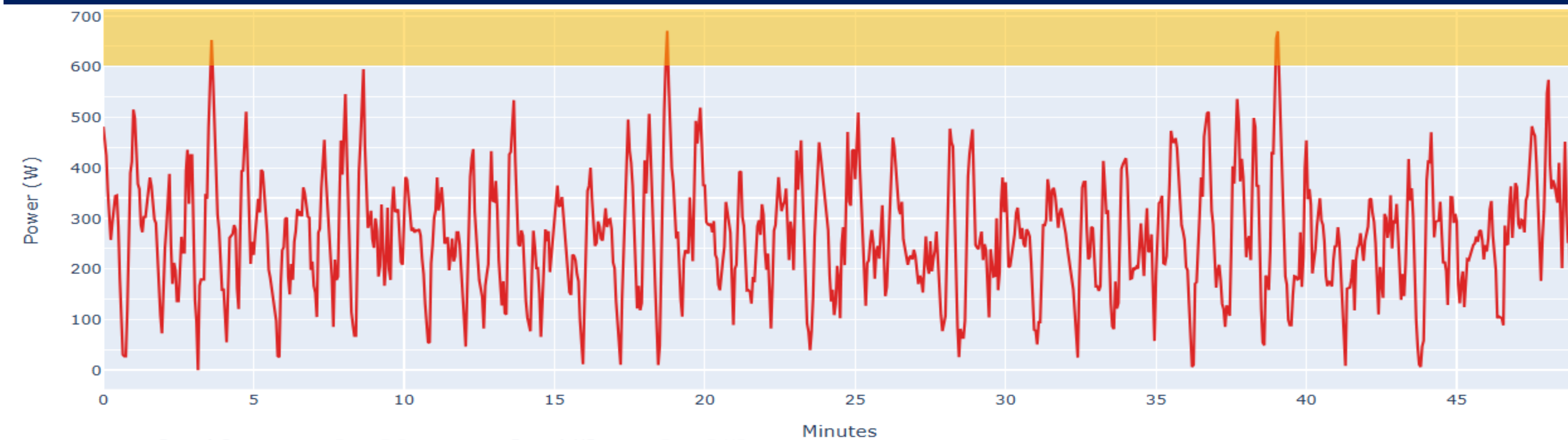
Bike Metrics

Average Power	259 W
Average Power w/o 0's	290 W
Normalized Power	279 W
Variability Index	1.08

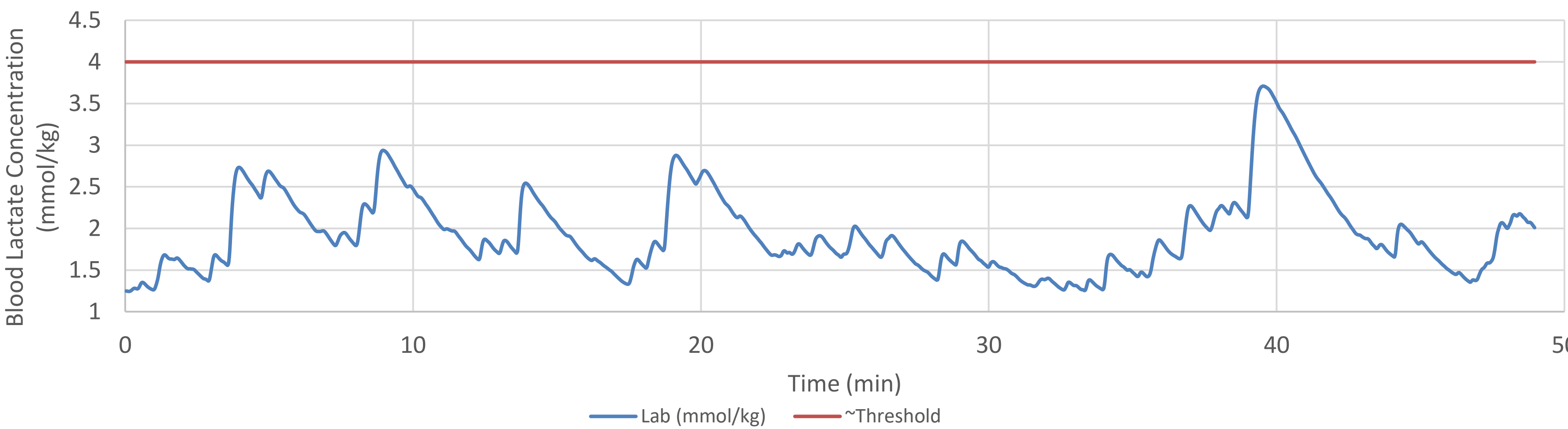
Braxton Legg Yokohama Bike

- As a taller athlete, Braxton is a bit heavier, but also has the highest threshold among the three. Despite an ability to hold higher power, Braxton minimized his surges over 600W to only 3 times during the ride. This kept his lactate below 4 mmol/kg for the whole effort.

Raw Power File



Simulated Blood Lactate Concentration



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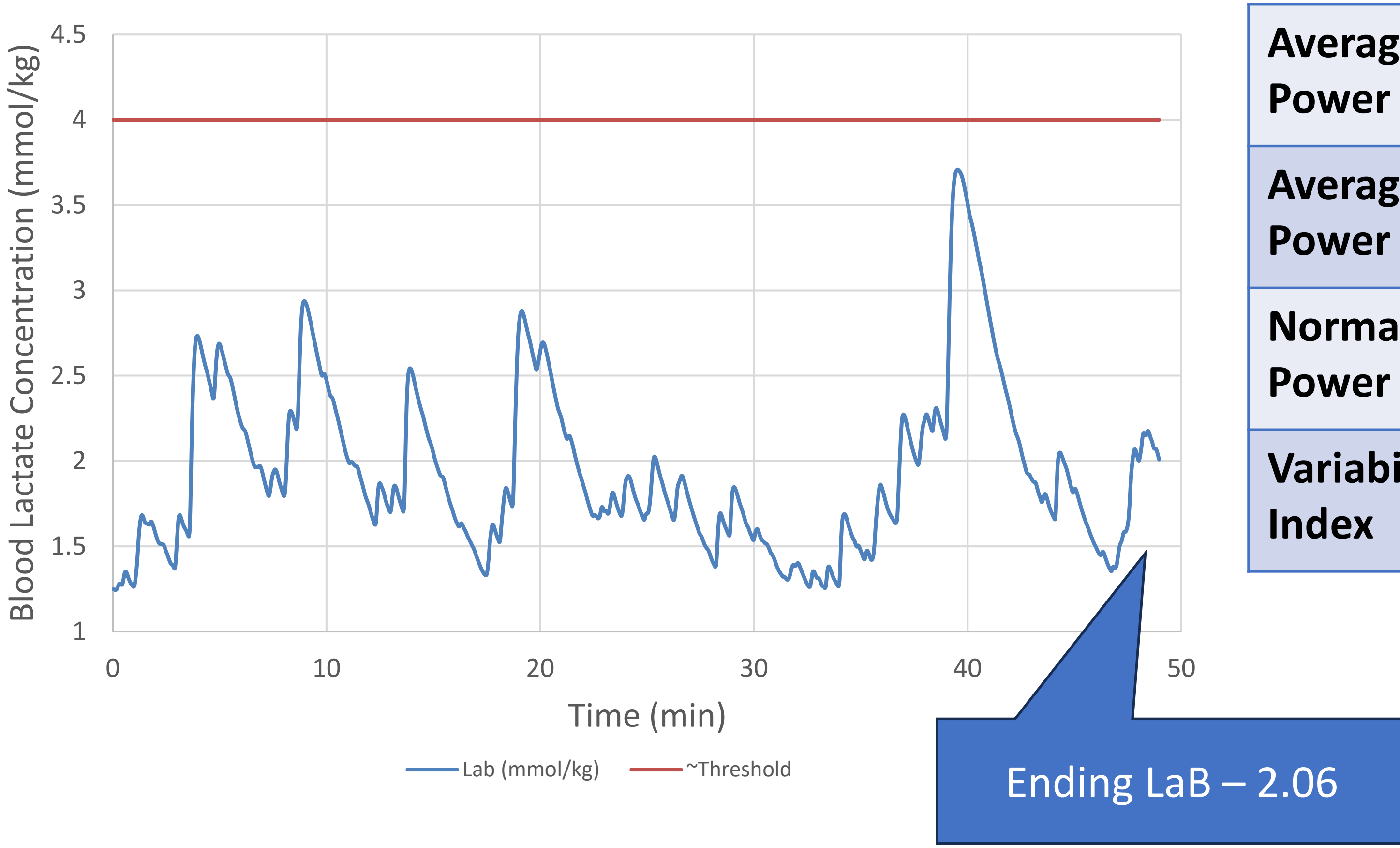
Braxton Legg Yokohama Bike

- Braxton also rode a mostly consistent effort with a VI of 1.09, keeping any surges to a minimum and riding a large portion of the ride near his maximum lactate clearance zone
- He finished the ride with lactate concentration of 2.06 and went on for his best run performance to date

Simulated Blood Lactate Concentration



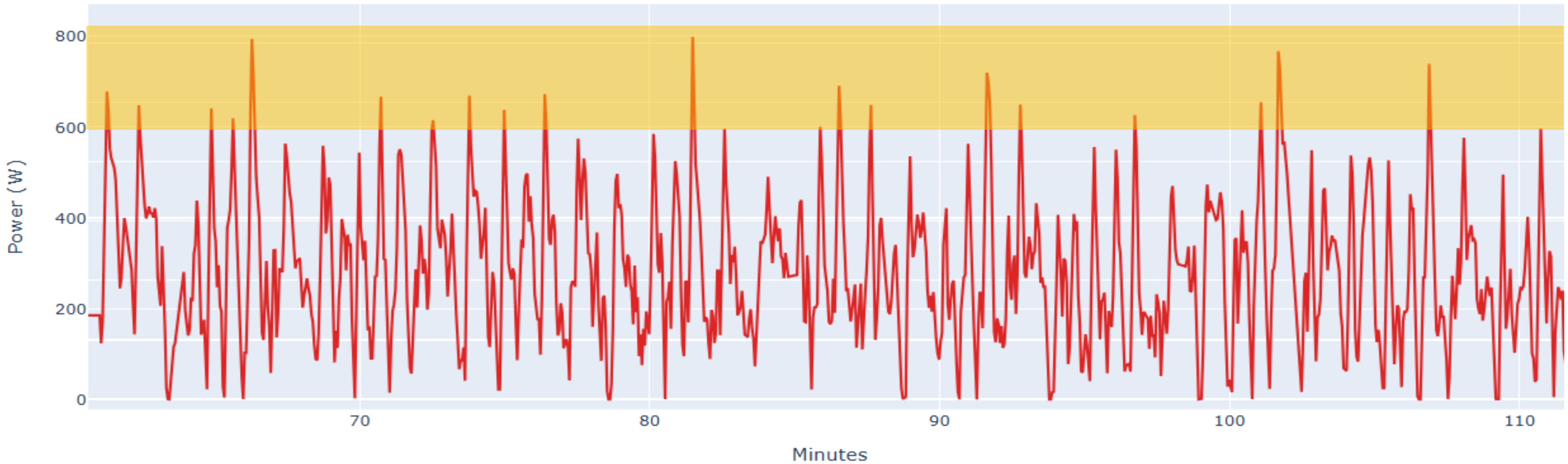
Bike Metrics

Average Power	**267 W**
Average Power w/o 0's	**307 W**
Normalized Power	**289 W**
Variability Index	**1.09**

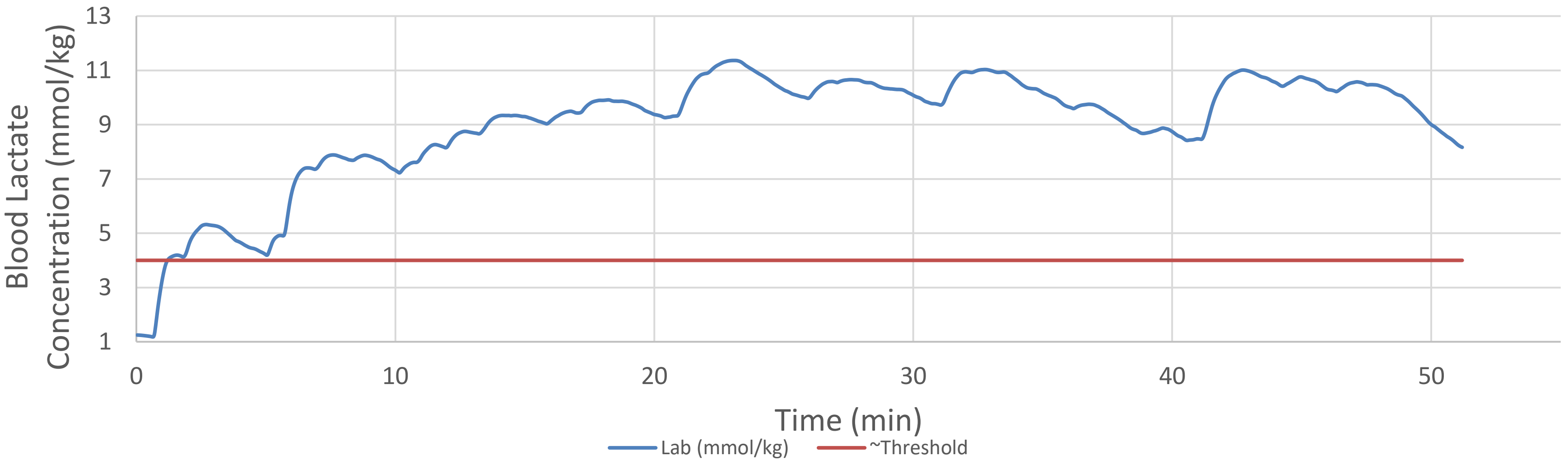
Reese Vannerson Yokohama Bike

- Reese spent significant time over 600 watts with several surges ~800 watts. This caused his lactate to rise significantly above threshold. Additionally, his bike positioning kept him from being able to ride his maximum lactate clearance wattage, keeping lactate high

Raw Power File



Simulated Blood Lactate Concentration

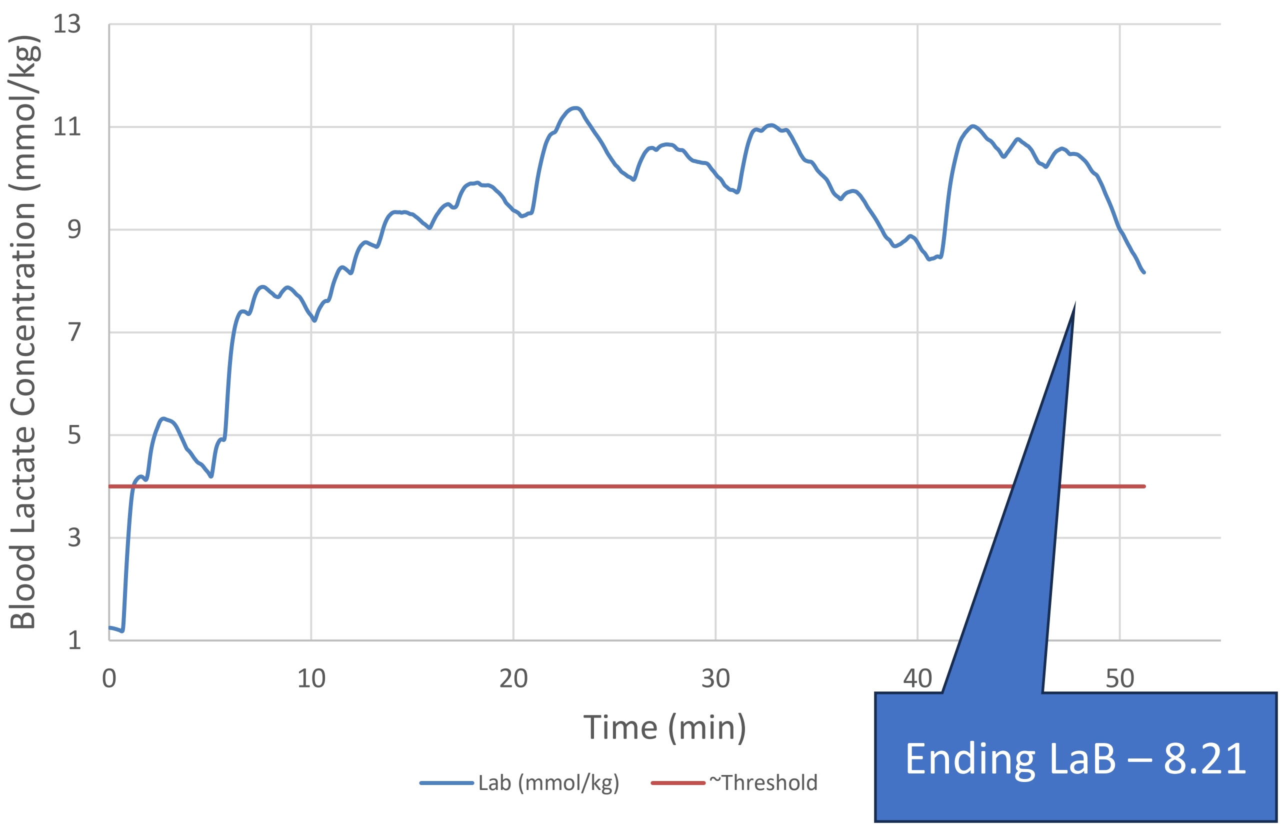


Note: Power file shown with 2s smoothing for better visibility

Reese Vannerson Yokohama Bike

- Despite a similar average power as Blake and Braxton, Reese had a higher Variability Index of 1.14 demonstrating the surges within his ride
- With an ending blood lactate value of 8.21, Reese was significantly over threshold beginning the run

Simulated Blood Lactate Concentration



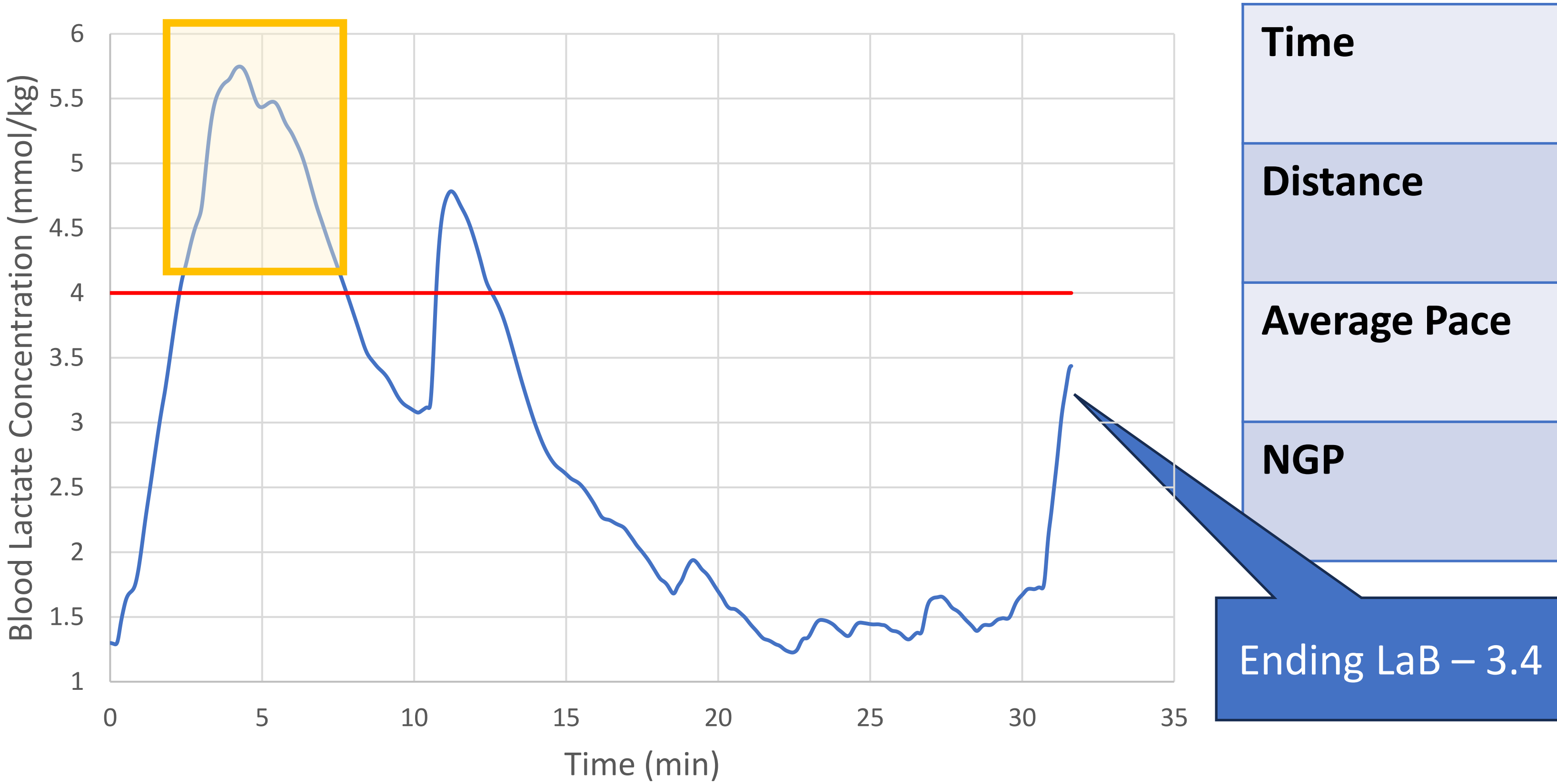
Bike Metrics

Average Power	**274 W**
Average Power w/o 0's	**314 W**
Normalized Power	**312 W**
Variability Index	**1.14**

Blake Bullard Yokohama Run

- Using Blake's Garmin watch we can simulate Blood Lactate using velocity as a proxy for the demands of the effort.
- Potential limitations of velocity may include lack of air resistance, elevation, and other environmental factors in the race that effect run speed but not necessarily power (effort).

Simulated Blood Lactate Concentration



Run Metrics

Time	**32:12**
Distance	**6.37**
Average Pace	**4:57**
NGP	**4:58**